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Continuous development and maintenance of an operational safety-critical system

A practical assurance model for live, continuously evolving railway systems

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Welcome



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Why this case study matters now

How can assurance keep pace with evolving capability? This case study describes an operating model to do just that.

1 THE OPERATING REALITY

24/7 automated freight; safety, availability and reliability all matter.

2 THE DELIVERY PRESSURE

Up to eight planned deployments per year, including novel and modified functions.

3 THE ASSURANCE RESPONSE

Tailor the lifecycle while preserving governance, traceability, independence and evidence.

Agility is safe only when assurance is designed into the delivery cadence.

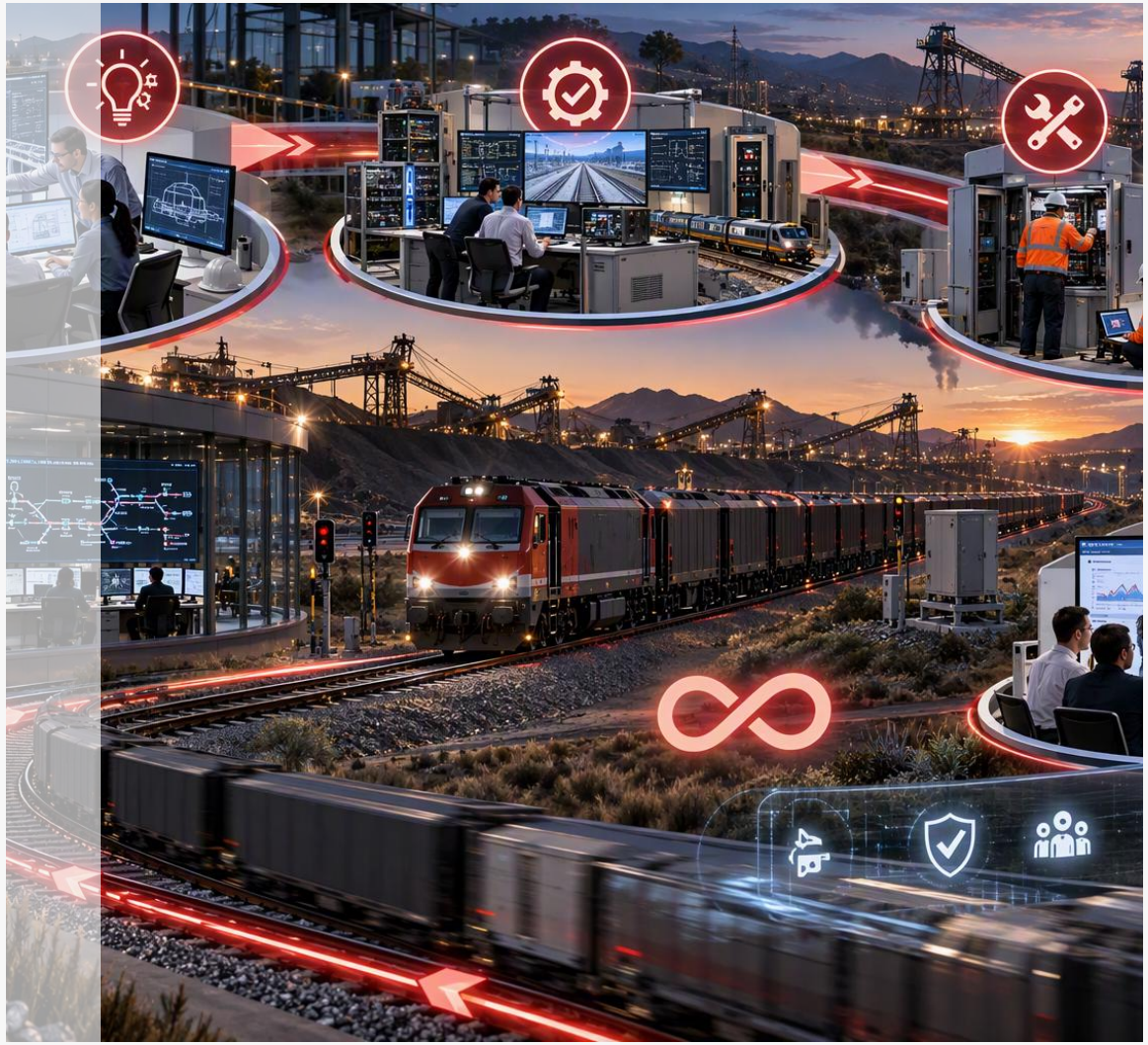
The Problem: The lifecycle does not end at commissioning

Traditional assumption

1. Concept
2. Definition
3. Implementation
4. Deployment
5. **Maintenance**

Operational reality

1. **Maintain**
2. **Enhance**
3. **Assure**
4. **Deploy**
5. **Learn**
6. **Repeat**



The system becomes a continuously evolving product, while remaining a continuously operating service.

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Case study at a glance

24/7	SIL 4	UP TO 8	2,000+	700+
live operation	rated automated safety systems	planned deployments each year	application conditions managed	superseded conditions cancelled

The non-negotiables

SAFE Every change analysed, controlled and evidenced.

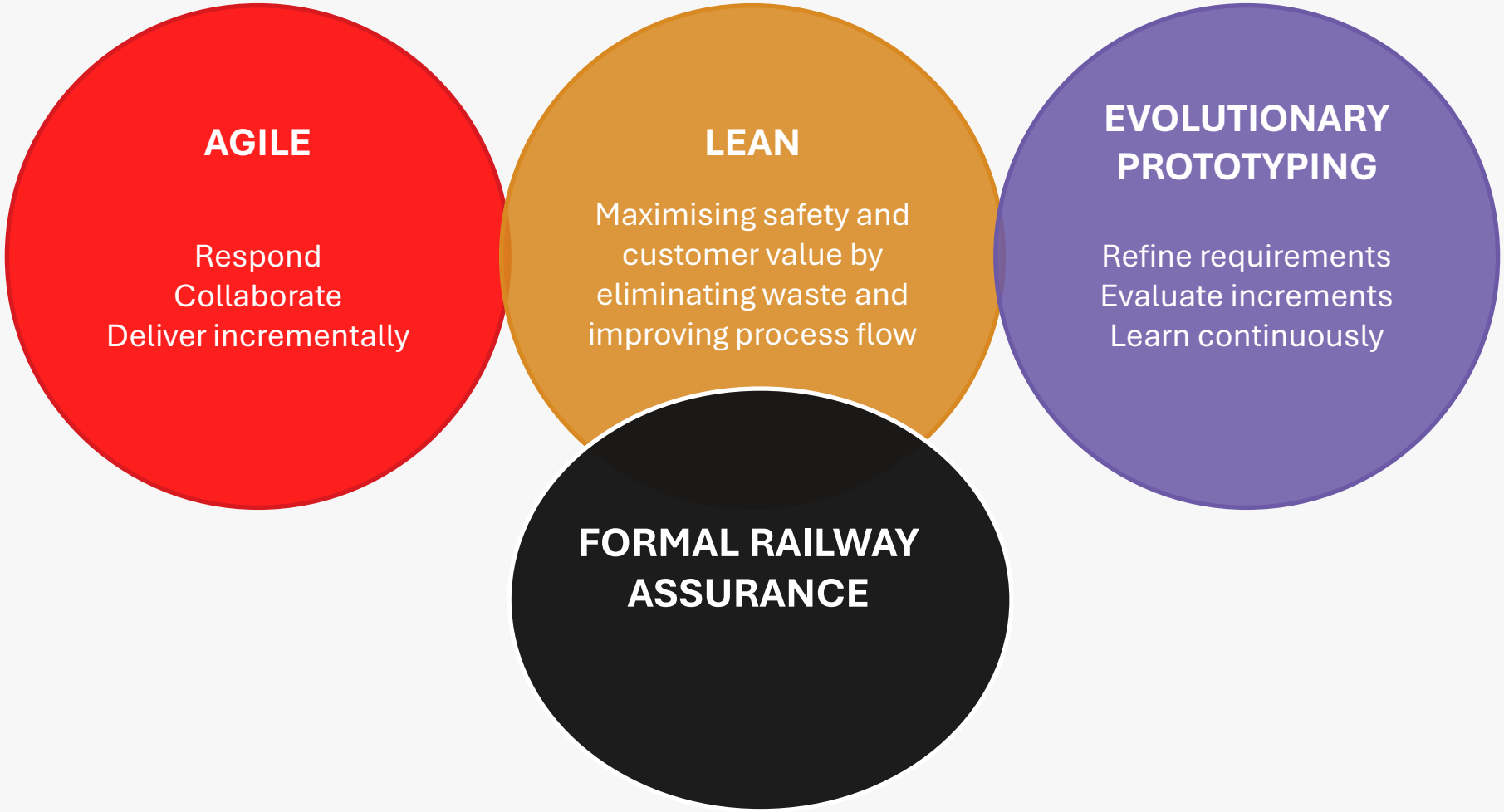
AVAILABLE Downtime minimised; backwards compatibility preferred.

RELIABLE Operational performance and defect priorities remain visible.

Design implication

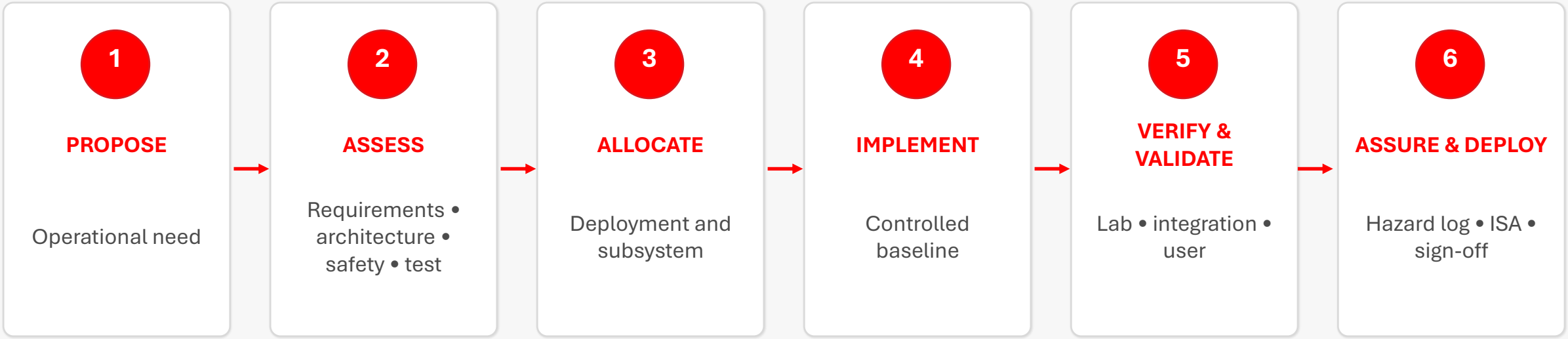
The deployment process must be treated as part of the safety architecture.

A hybrid model: evolve fast, assure continuously



Configuration control • Hazard management • V&V • ISA independence • Traceability

Every change passes through the same level of checks



DEFER, DO NOT DILUTE • If analysis and assurance cannot be completed properly, move the change to a later deployment.

Principle 1:Engage the operator before requirements harden



1 **ROADMAP**

Agree change intent and sequencing up to a year ahead.

2 **DESIGN DEFINITION**

Jointly review requirements, architecture, safety, testing and human factors.

3 **EARLY EXPOSURE**

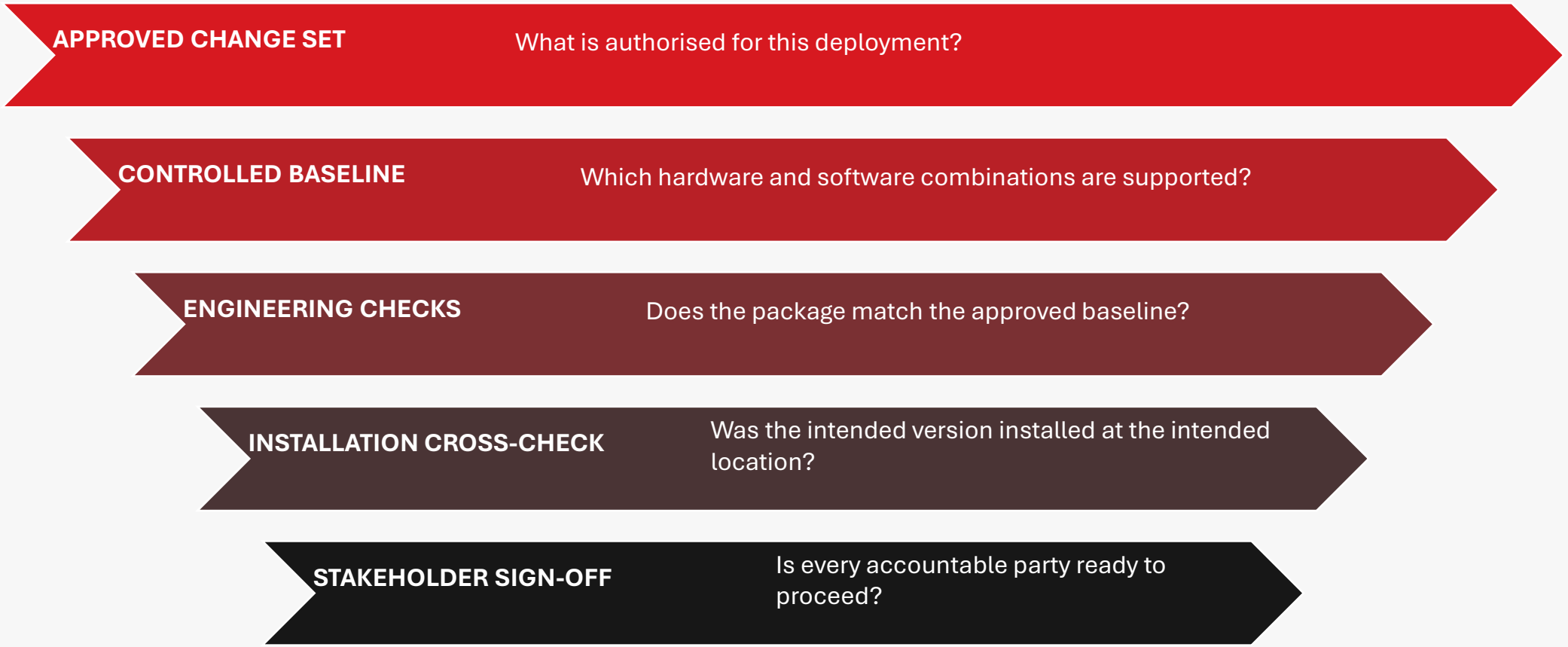
Use laboratory demonstrations and controlled site testing.

4 **ACCEPTANCE**

Validate not only function, but operational fit and expected behaviour.

Outcome: requirements evolve with evidence, not after delivery.

Principle 2:Configuration management is the deployment backbone

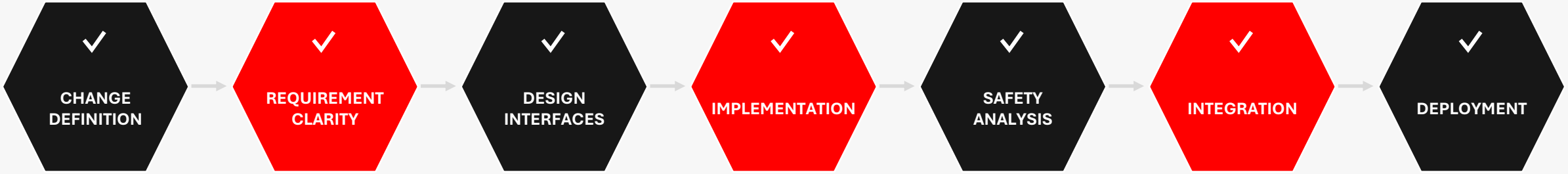


Frequent delivery increases the value of control, not the tolerance for ambiguity.

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Principle 3:Verification is continuous, not a final gate

“There is no time to correct later in the lifecycle.”



Verification is built into the work: subsystem and system teams communicate early, expose assumptions and confirm interfaces before integration becomes the discovery mechanism.

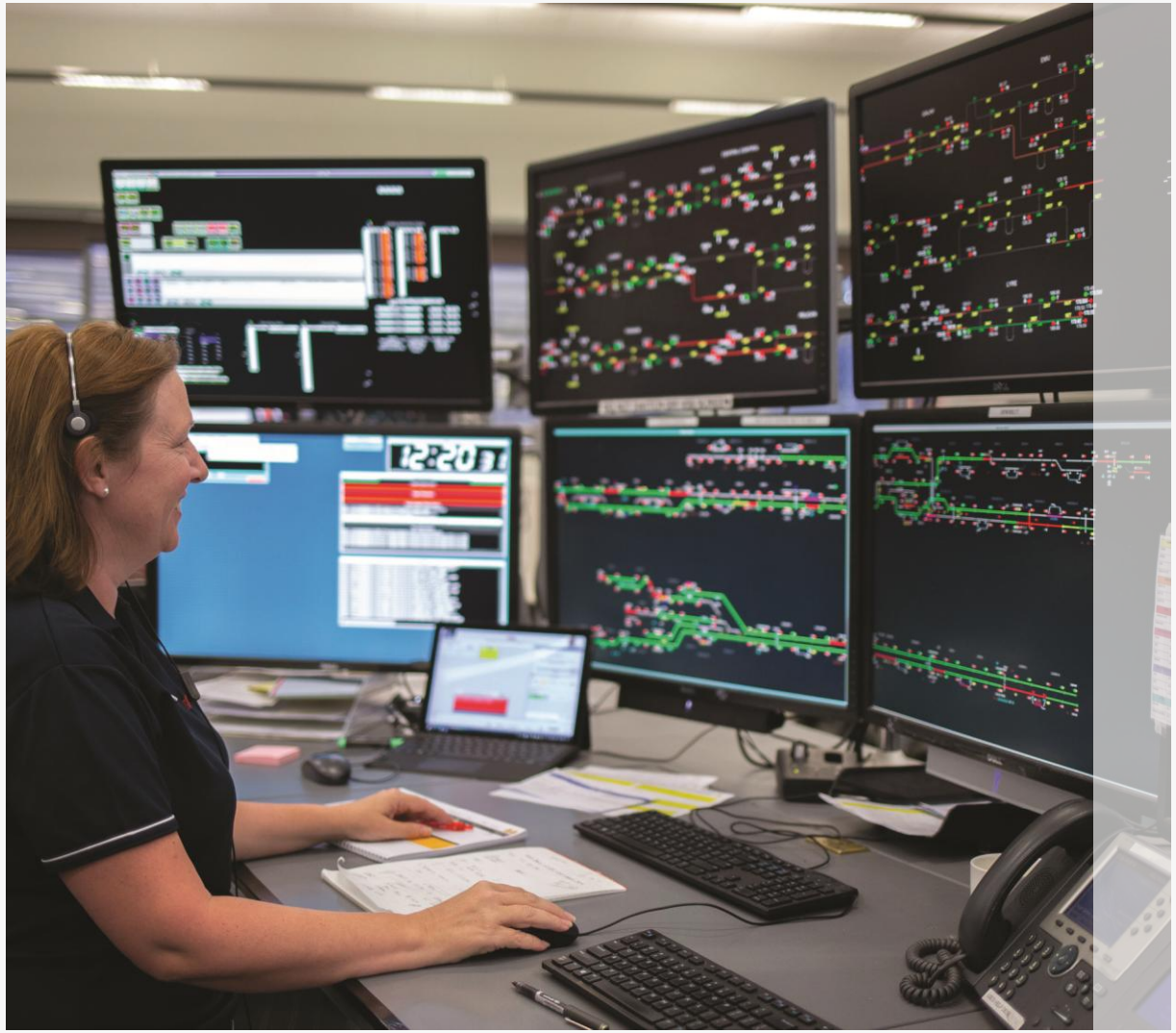
Principle 4: One live hazard log connects change to safety case



CHANGE ASSESSMENT	Preliminary safety and operational impact
↓	
TARGETED ANALYSES	OSHA • FFA • IHA, as applicable
↓	
MASTER HAZARD LOG	Live parent; controls and application conditions
↓	
TRACEABILITY	Controls → requirements → V&V evidence
↓	
SAFETY CASE	SFAIRP argument updated with deployment evidence

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Principle 5: Validation asks two different questions



1 DID IT WORK?

Functional and technical conformance

2 DID IT WORK AS NEEDED?

Operational value, human factors and user expectation

Simulation → early integration → system testing → controlled site testing → operational feedback

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Principle 6: Application conditions are active risk controls

BEHAVIOUR

Highlight changed function behaviour

BASELINES

Identify newly supported configurations



RESIDUAL RISK

Communicate procedural mitigations

DEFECTS

Bridge safety or high operational impact

700+ superseded conditions cancelled as new updates superseded them

Principle 7: Treat the ISA as an early stakeholder, while preserving independence

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1

EARLY VISIBILITY

Share deployment roadmap and planned safety-related changes.

→

2

OPEN ISSUES

Raise concerns, uncertainties and proposed responses without delay.

→

3

TIME TO UNDERSTAND

Provide evidence early enough for meaningful challenge.

→

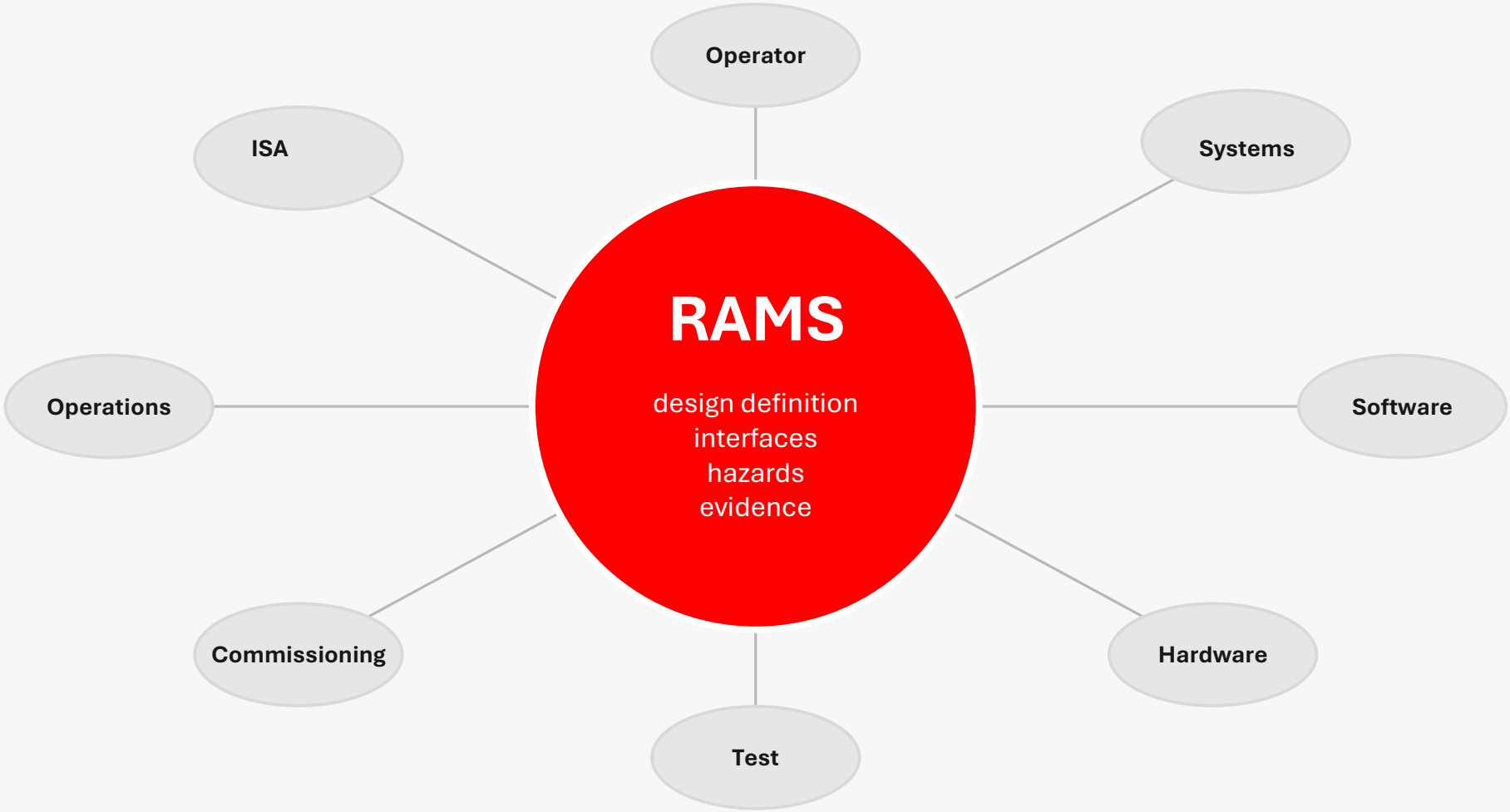
4

NO LATE SURPRISES

Resolve comments before they become deployment blockers.

Transparency accelerates assurance because challenge happens while options are still open.

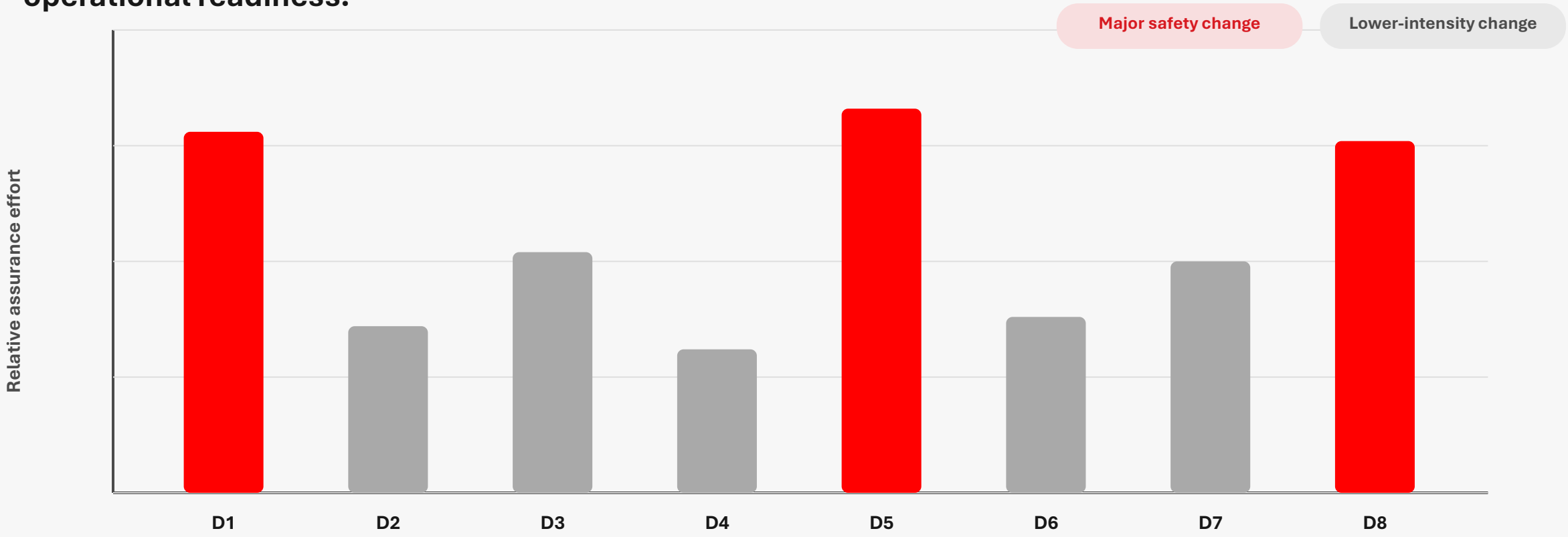
Principle 8: RAMS is embedded in the system integration process



One team ≠ one role. Collaboration must coexist with defined accountability and required independence.

Principle 9: System assurance workload

A roadmap is not only a delivery plan. It is a control for assurance capacity, integration risk and operational readiness.



Levelled flow protects decision quality, review depth and sustainable team performance.

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How the Railway Processes align with Agile, Lean & Evolutionary prototyping

Railway Processes	AGILE	LEAN	EVOLUTIONARY PROTOTYPING
Early engagement with operator	Continuous collaboration	Customer value	Active user involvement
Strong Configuration management	Reliable delivery	Avoiding waste	Control evolution
Embedded Verification	Continuous testing	Preventing defects	Verification checks
Hazard Management	Technical excellence	Eliminating waste	Incremental Delivery
System Validation	Customer-value	Delivering real customer value	Incremental development
Robust application conditions	Enabling safe incremental changes	Avoiding duplicated analysis	Continuous customer involvement
ISA engagement	Early feedback	Avoid blockers	Continuous evaluation
Cross Functional cooperation	Collaborative culture	Reducing organisational barriers	Active user involvement
Workload Management	Precise scope (achievable in sprint)	Optimisation principles	Improves overall process flow.

No method is applied mechanically. The project adopts principles that strengthen safe, repeatable delivery.

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What the case study teaches

01

TAILOR THE PROCESS, NOT THE SAFETY OBLIGATION

02

MAKE EVIDENCE A CONTINUOUS FLOW

03

INVOLVE USERS AND ASSESSORS EARLY

04

DEFER WORK WHEN ASSURANCE CANNOT FIT

05

DESIGN THE CADENCE AROUND CAPACITY

Things will still go wrong. A mature process shortens the path from detection to impact assessment, mitigation and safe recovery.

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Future Direction: turn assurance evidence into a digital thread



MODEL-BASED ENGINEERING
Link system models, hazards, interfaces and requirements.

AUTOMATED SCENARIOS
Execute repeatable operational and failure scenarios continuously.

TRACEABILITY CHECKS
Detect missing or broken evidence links before review.

EVIDENCE GENERATION
Automate collection and packaging of verification results.

OPERATIONAL ANALYTICS
Use events, trends, performance and user feedback to identify emerging risk.

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Conclusion

Operator engagement

Configuration Control

Embedded V&V

Hazard Management

Validation

Application conditions

ISA engagement

RAMS in System integration

System assurance workload

The takeaway

Agile responsiveness, Lean flow and evolutionary learning can coexist with formal railway safety assurance.

When assurance is embedded in every deployment, continuous change becomes governable.

Questions & Discussion

Dr Andrew Hussey | Graziano Marino | Maria Hill | Mohan Murari